SHORTHAND THEMES FOR BOOKS

As early as I can remember, I wanted to be a musician, and I became a biologist.

Are living creatures like pieces of chemical music? (The reproduction of individual organisms - like different concerts of the same piece of music? Biological species - different pieces of music?)
Even a single bacterium is a damn good piece of molecular homeostasis/harmony/music

Biology feeds on material order (food, selected for energy and electrolyte provision) to grow cells and bodies.
Culture feeds on informational order (selected emotionally for harmony and rationally for meaning) to grow more culture(s)

Biological species (and individuals within) compete <u>brutally</u> - but also in solidary packs
Cultural species compete <u>cruelly</u> (in large linguistic, ethnic, national or religious packs)

Why do we do what we do?

To generate our emotions and "feed" on emotions as diagnostics, motivator and driving force of human life.

What is emotion?

- Ultimate holistic/integrative diagnostics of the level of system's harmony, and an ultimate selection criterion? We keep what "feels good", even in science!
  Earning money (energy?) feels damn good - so, we have capitalism.

Compare with harmony and disharmony in music.
What did Stravinsky & others want to feel? Music of "losers"?
How about emotional engineering (Hitler, populism, marketing, publicity…)

3D Printing of Human Beings as pedagogical exercise and ultimate trans-humanism

If we could 3D print a human being, that would be the evidence that we know what is life and who we are.

Is embryogenesis a life-creating 3D printing process preprogramed in DNA? But, how?
Difference - engineering vs evolution:
-**Engineering approach:** preparing beforehand bits and pieces and modules, and then composing them in a system that functions strictly predictably.
-**Embryogenesis approach**: growing bits and pieces and modules in right (pre-composed) place and time into a system that plays the music of functions - homeostasis/harmony - but not absolutely and strictly predictable
-**Error correction mechanisms** in embryogenesis (Cell apoptosis? Spontaneous abortion?) - who/what sees the error and decides about early selective death (is it also some kind of "emotion" as harmony-detector?)
How is growth (time sequence of constructive steps) of organism preprogramed in DNA?

Music & mathematics.

How would we 3D print human beings? As engineers or as evolutionists?
Our intellectual limitation is that we try to think and do things rationally, like engineers, not like evolution.
Yet, our elementary thinking process is like super-fast evolutionary process (selection of few from many options). With exception of inborn reflexes, nothing is said in advance - it must be a process, a "growth", each time..) - but the selection is emotional (because it is holistic-integrative. Rational thinking is analytical and therefore has low synthetic, integrative capacity)

A cultural problem: Western culture handicapped itself by not allowing emotions the "citizenship" of humanity (we prefer reason, but analytical reasoning deals only with measurable immediate efficacy?). Emotion of experienced, knowledgeable, person integrates knowledge into wisdom and wisdom serves both present (immediacy) and future. We should develop arts of cultivating emotions!

*Edward de Bono said that love (*emotion?*) is too unstable currency upon which to base our society; he proposed respect as the basis (but respect is also emotion!)*

So, what kind of human would we 3D prints?
This is of course a purely philosophical question, but useful to define what we really want, now.
I guess that human 3D printers should be rather like evolutionists than like engineers - thus, pretty much what we already have: babies who know nothing and can learn everything! (Maybe that a higher material quality of the printed human "hardware" would be useful to live longer and do more?)
So, one should print babies not adults (just like we already have with our reproduction) to avoid real cloning of clones who think identically.
To stop evolution by cloning adults would be a mortal sin.

Humanism as Science:
the structure of harmony in music, health, knowledge, emotions and ethics.

Each complex structure emerges by one of two strategies: (i) actively and "spontaneously" by autonomous growth according to an intrinsic pre-programed plan, like for living organisms, or (ii) passively by being constructed according to the plan of an engineer external to the structure, like in man-made technologies. It is conceivable that a class of futuristic robots could emerge from a mixture of two strategies.

Construction by growth is particularly interesting nature's evolutionary way to make also human beings; it proceeds by the paradigmatic branching growth of trees. When we map growth of living creature from the first cell (zygote) to finished organism, it grows like a tree! The final product (an animal) takes forms unlike trees but grows by bifurcation of successive cell divisions that we can see macroscopically as the branching structures of many trees.

To tackle the structure of knowledge we can first appreciate that the brain grows hundreds of billions of neurones also like a tree. But, unlike the usual trees with well separated branches, neuronal branches in the brain establish also horizontal inter-neuronal connections resulting in 3D networks.

Questions:
Is the emerging knowledge logically structured - as it grows - like a tree, or rather a tree of trees that are both vertical and horizontal?
Can we learn about this from the structure of mistakes we make by thinking? (This is exactly how the knowledge in genetics grew: generate mistakes (mutations) and study their consequences)*.
Which, presumably old, branches constitute the "common sense" thinking versus recently added branches of specialised knowledge?

**This is why studying diseases is teaching more about normal functions than about the origin of diseases. Hence, studying robust health would teach about diseases (why they don't occur).*

((I believe that today's biomedical research is on a mistaken path (took wrong unproductive cognitive branches). Will further continuation of such research lead to getting increasingly lost (going away from solutions)? Is it smarter to descend back to the solid basic branches, get common sense, and chose alternative upwards branch?))

The cognitive value of emotions

Traditionally, feelings, emotions are feared by serious scientists as the irrational enemy biasing their rational thinking. But, is this an over-simplification of the role of emotions in cognition? Can it be that emotion is our superior cognitive tool diagnosing holistically, in the global, what's smart and wise and what's stupid (sometimes intelligent but fatal)? Emotion might have evolved to foster the ultimate good - group survival - thanks to the aesthetics and harmony and ethics?

We call geniuses people who arrive to the astonishing new knowledge or aesthetics by a quantum leap thanks their talent (the talent is the "IQ" of the intuitive/emotional - both are holistic and apparently irrational harmony detectors. This way, geniuses arrive to the success by taking immensely efficient "short cuts" in the jungle of branches in the cognitive tree.

Separate well positive and pleasant from negative painful emotions - they are diagnostic means for the quality and value of what we are doing:

Positive emotion means harmony and sais "well done, keep doing it"
Negative emotion means disharmony and sais "don't do that anymore"

Medicine: Study of health, not disease!

Health is beautiful, desired, interesting and fun;
Disease is ugly, undesired, boring and sad
Why study what we don't want (disease) instead of what we want (health)?

Now, this is all a nice subject for ARTSCIENCE!